Which Zero-Shot Signals Predict Low-Bit Quantization Fragility? A Cross-Dataset Audit on 46 Pretrained Vision Models

First Author[1][0000-0000-0000-0000], Second Author[1][0000-0000-0000-0000], Third Author[1][0000-0000-0000-0000], Fourth Author[1][0000-0000-0000-0000], and Fifth Author[1][0000-0000-0000-0000]

Department of Information Technology, Chaitanya Bharathi Institute of Technology, Hyderabad, India
first.author@example.org

Abstract. Typically, selecting which pretrained backbone to deploy under 4-bit post-training quantization (PTQ) is done by quantizing and evaluating every candidate. We ask whether one forward pass on 32 unlabeled images can predict how much accuracy a model will lose. We propose Amos, a ratio of layer-output activation kurtosis to the normalized top-1/top-2 logit margin, and audit it against 11 additional signals (zero-cost NAS proxies, Hessian- and weight-statistics-based scores, model size, and FP32 accuracy) across 46 pretrained models: 16 CIFAR-100 CNNs and 30 ImageNet models from 10 architecture families. The models were evaluated on CIFAR-100 and ImageNetV2 using four bit-width settings and three calibration seeds. On CIFAR-100, Amos tracks W4A4 degradation (Spearman $\rho = 0.87$, 95% CI $[0.62, 0.96]$). The relationship is robust to parameter count and weight range and remains stable after controlling for architecture family and multiple testing. A prespecified transfer test shows that this ratio does not hold on ImageNetV2 ($\rho = -0.47$). Decomposing the score helps explain why: the outlier component retains its sign and passes the size and weight-range controls on both datasets ($\rho = +0.57$ and $+0.65$), while the margin component reverses. Parameter count is the most consistent predictor of W4A8 loss across both datasets ($\rho = -0.90$ and -0.68), while ranking by FP32 accuracy mis-selects models at W4A4. The signals cost 0.06–0.2 s per model, a median of 175–201 times less time than a three-seed PTQ evaluation. We provide an audit protocol and code to validate inexpensive quantization proxies across datasets.

Keywords: Post-training quantization · Zero-cost proxies · Model selection · Activation outliers · Reproducibility.

1 Introduction

When deploying a vision model on an edge device, it is often a matter of selecting among a variety of pretrained backbones and compressing the winner using post-training quantization (PTQ) [1, 2]. Under aggressive settings, such as 4-bit weights with 4-bit activations (W4A4), some architectures lose only a few points of accuracy while others collapse to chance (Sect. 5.1). FP32 accuracy does not distinguish between them, so the

practical recipe is to quantize in a loop that is repeated for each bit-width and calibration subset, with every candidate calibrated and evaluated.

In neural architecture search (NAS), zero-cost proxies avoid this loop: they score an architecture in a single pass (forward or backward) at initialization [3–6]. These proxies target the trainability of the FP32 model, not quantization behavior. Quantization-specific signals exist but are used differently: Hessian-based sensitivity allocates bits within a given model [8, 9], and training-free proxies drive mixed-precision search [10]. Kurtosis-based objectives suppress activation outliers during training or rotation [11]. Whether a cheap, label-free signal can rank different pretrained architectures by their PTQ fragility, and whether such a ranking is transferable across datasets, is not often tested. Spurious proxy success on small model zoos is easy to overlook.

In this paper, we study that question as an audit. We propose Amos (Analytic Margin–Outlier Sensitivity), calculated from one forward pass of an FP32 model on 32 unlabeled images. It is evaluated against 11 reference signals on 46 pretrained models, with a fully simulated PTQ pipeline as ground truth. The evaluation is based on three controls: within-family correlation, which is typically missing from studies that pool correlations across model zoos; partial correlation controlling for the number of parameters and weight range; and leave-one-family-out prediction. Our contributions are:

1. **An audit protocol and benchmark.** We evaluate 46 models (16 CIFAR-100 CNNs and 30 ImageNet models from 10 families, tested on ImageNetV2) across four bit-width settings, three calibration seeds, and 12 zero-shot signals. The protocol includes bootstrap intervals, family-adjusted and partial correlations, held-out-family prediction, selection regret, and multiple-testing correction.
2. **Good in-distribution performance.** Amos is correlated with the W4A4 accuracy drop on CIFAR-100 at $\rho = +0.87$ (95% CI $[0.62, 0.96]$). It keeps $\rho \geq +0.77$ when any family of architectures is held out, and the result remains at $+0.83$ when the number of parameters or the weight range is controlled.
3. **A pre-declared transfer test and its explanation.** The ratio is not transferable to ImageNetV2 ($\rho = -0.47$). When the ratio is broken down into its two parts, the outlier term transfers in sign and survives both controls ($\rho = +0.57$ and $+0.65$ pooled; $+0.89$ and $+0.67$ given parameter count). The margin term is reversed.
4. **Consistent cross-dataset findings.** The number of parameters is a consistent predictor of W4A8 loss across both datasets ($\rho = -0.90$ and -0.68). Ranking by FP32 accuracy mis-selects at W4A4, performing worse than random selection on ImageNetV2. The measurements are cheap and stable: 0.06–0.2 s per model, a median of 175–201× faster than three-seed PTQ. The rankings are consistent across probe batches ($\rho = 0.95$–0.99) and across probe sizes starting from $N = 8$ images.

Code, notebooks and result tables are available at `[REPOSITORY URL TO BE ADDED]` (see Sect. 4).

2 Related Work

Zero-cost proxies. The overlap of activation patterns on a batch is scored by NASWOT [3]; SynFlow measures data-free synaptic saliency [4]. Gradient-norm and related proxies were benchmarked for lightweight NAS [5], and Zen-NAS ranks architectures by

expressivity [6]. All target FP32 performance. Lukasik et al. assess zero-cost proxies for robustness [7]. We ask the analogous question for quantization instead of robustness.

Quantization sensitivity. HAWQ and HAWQ-V2 allocate precision across layers based on Hessian information [8, 9]; EMQ evolves training-free proxies for mixed-precision search [10]. These methods address bit allocation within a model. We consider a complementary setting: pre-ranking various pretrained models without quantization.

Activation outliers and weight statistics. Heavy-tailed activations cause quantization ranges to become wider. They are important for large-model quantization [13, 14] and are minimized during training using kurtosis regularization [11]. Cross-layer range imbalance in the weights motivates data-free quantization [12]. As data-free reference signals, we use the weight-range ratio and weight kurtosis.

3 Method

3.1 Problem setting

Let M be a pretrained FP32 classifier and $\mathrm{Acc}_b(M)$ its top-1 accuracy after PTQ at bit-width setting $b \in \{\mathrm{W8A8}, \mathrm{W4A8}, \mathrm{W6A6}, \mathrm{W4A4}\}$ (WwAa: w-bit weights, a-bit activations). The target is the accuracy drop $\Delta_b(M) = \mathrm{Acc}_{\mathrm{FP32}}(M) - \mathrm{Acc}_b(M)$ in percentage points (pp), averaged over calibration subsets. A zero-shot signal $s(M)$ is computed without quantizing M and without labels; we assess how well it ranks a zoo of models by Δ_b and by the quantized accuracy.

3.2 AMOS: Analytic Margin–Outlier Sensitivity

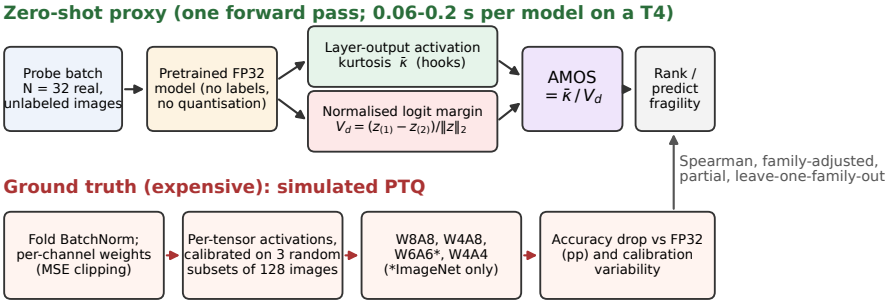
Fig. 1 summarizes the computation. For a probe batch X of $N = 32$ unlabeled images, a forward pass with hooks on every convolution and linear layer yields the output activation tensor A_l of layer $l = 1, \ldots, L$. Its kurtosis is

$$\kappa_l = \frac{\mathrm{E}\big[(A_l - \mu_l)^4\big]}{\big(\mathrm{E}\big[(A_l - \mu_l)^2\big]\big)^2}, \tag{1}$$

with expectations over all elements of the tensor. The first and last layers are excluded because they are kept at 8 bit in our PTQ protocol, and the remaining layers are averaged, $\bar{\kappa} = \frac{1}{L-2}\sum_{l=2}^{L-1} \kappa_l$. The logits z_i give the normalized decision margin

$$V_d = \frac{1}{N}\sum_{i=1}^{N} \frac{z_{i,(1)} - z_{i,(2)}}{\|z_i\|_2 + \epsilon}, \qquad \mathrm{AMOS}(M) = \frac{\bar{\kappa}}{V_d}, \tag{2}$$

where $z_{i,(1)} \geq z_{i,(2)}$ are the two largest logits and $\epsilon = 10^{-6}$. The intuition is that heavy-tailed activations force wide clipping ranges that waste quantization levels, while a small margin leaves a prediction easy to flip. We call $\bar{\kappa}$ the *outlier term* and $1/V_d$ the *margin term*. A variant, AMOS-in, measures κ_l on the *inputs* of each layer, i.e. on the tensors that the activation quantizer actually sees. We report both. We also study the two terms separately (*output-κ only* and *1/margin only*) and alternative aggregations over layers (mean, geometric mean, median, maximum, parameter- and depth-weighted).



Flow diagram in two rows. Top row: 32 unlabeled images enter a pretrained FP32 model; layer-output activation kurtosis and normalized logit margin are computed in parallel and combined into the AMOS score, which is used to rank or predict fragility. Bottom row: the ground-truth path folds batch normalization, quantizes weights and activations on three calibration subsets, runs four bit-width settings and records the accuracy drop.

Fig. 1. Overview. The zero-shot signal needs one forward pass of the FP32 model on 32 unlabeled images. The ground truth requires BN folding, weight and activation quantization, three calibration subsets and a full evaluation pass for every bit-width setting; the two are compared with the statistics of Sect. 3.5.

3.3 Reference signals

We compare against: NASWOT [3]; SynFlow [4] (batch-norm layers in evaluation mode); gradient norm [5]; a HAWQ-style score [8], i.e. $\sum_l \mathrm{Tr}(H_l)/n_l \cdot \|Q_4(W_l) - W_l\|^2$ with a Hutchinson trace estimate (it needs double back-propagation and is undefined for layers whose second derivative is unavailable, such as hard-sigmoid gates); the weight-range ratio (cross-channel range dispersion of BN-folded weights) and weight kurtosis [12, 11]; parameter count; and FP32 accuracy on the evaluation set. Signals that cannot be computed for a model (for example NASWOT on some architectures, or the HAWQ-style score where double back-propagation is unsupported) are omitted from that signal's statistics.

3.4 Ground-truth PTQ simulator

All ground-truth data is provided by a single simulator to ensure that models are handled equally. BatchNorm is folded into the preceding convolution (numerically verified for each model). Weights are quantized symmetrically per output channel, with an MSE-optimal clipping search for fewer than 8 bits. The input of each convolution and linear layer is quantized per tensor with an asymmetric uniform quantizer. Its range is the 99.99th percentile of the activations on a calibration subset of 128 images (processed in two batches of 64, with the range averaged over batches). The first and last layers remain at 8 bits. The attention matrix products, softmax, and normalization layers remain in floating point. Three random calibration subsets are used for each configuration, and the mean accuracy is reported.

3.5 Evaluation protocol

We report Spearman ρ and Kendall τ with 95% bootstrap intervals (2000 resamples of models). Because model zoos are clustered into architecture families, we add: (i) *family-adjusted* correlation, i.e. Pearson correlation of ranks demeaned within family; (ii) *partial* rank correlation given parameter count or the weight-range ratio; (iii) *drop-one-family* correlations; (iv) *leave-one-family-out* (LOFO) prediction of quantized accuracy by a ridge regression on FP32 accuracy and one signal, compared with predicting the training mean; (v) *selection regret@1*, i.e. the accuracy gap between the best model and the model picked by a leave-one-out regression; and (vi) a Holm correction over all variant-by-target tests of the ablation. We declared one confirmatory hypothesis before running the ImageNetV2 experiments:

> **H1.** The primary Amos score correlates positively with the W4A4 accuracy drop (secondary: W6A6).

CIFAR-100 is the discovery set; ImageNetV2 is the confirmation set.

4 Experimental Setup

Model zoos. The CIFAR-100 zoo has 16 pretrained CNNs (ResNet-20/32/44/56, VGG-11/13/16/19-BN, MobileNetV2 at four widths, and ShuffleNetV2 at four widths) from a public checkpoint repository [15]. Evaluation uses 10,000 test images, while probe and calibration images are drawn from the training split.

The ImageNet zoo has 30 pretrained models from the `timm` library [16], covering 10 model families: ResNet, VGG, MobileNetV2/V3, EfficientNet, MnasNet, GhostNet, RegNet, ConvNeXt, and ViT/DeiT (Table 1). The pretrained recipe of each model is recorded. The models are tested on ImageNetV2 matched-frequency [17], using 5,000 evaluation images and 1,000 images (one per class) for probes and calibration, with the sets kept disjoint. ImageNetV2 has the same label space as ImageNet [18] but has lower absolute accuracies than the original validation set; all drops are relative to the FP32 accuracy on the same images. Architectures are cited in [19–29].

Bit-width settings and signal computation. W8A8 (control), W4A8, and W4A4 are used for both datasets, with W6A6 added on ImageNetV2. Fourteen of the 30 ImageNet models collapse below 3% top-1 under W4A4 per-tensor PTQ, so W6A6 is the graded stress setting there. Amos is averaged over 10 probe batches of 32 images on both datasets; NASWOT, SynFlow and gradient norm are averaged over the first 3 batches, and the HAWQ-style score uses one batch (on ImageNetV2, gradient-based baselines use 16 images to fit in memory). The probe ablations use 3 batches. All timings were measured on a Tesla T4 GPU.

Reproducibility. All results can be regenerated with the released notebook, which checkpoints every unit of work and verifies cached results against the current data before use. The repository `[REPOSITORY URL TO BE ADDED]` contains the notebook, the per-model result tables behind every figure and table, the scripts that draw the figures, and the random seeds used for probes and calibration subsets.

Table 1. Model zoos. Parameters in millions; FP32 top-1 accuracy (%) on the respective evaluation set.

Dataset	Family	#Models	Params (M)	FP32 top-1
CIFAR-100	mobilenetv2	4	0.82–4.50	71.2–76.3
	resnet	4	0.28–0.86	68.8–72.6
	shufflenetv2	4	0.44–5.55	67.8–75.5
	vgg	4	9.80–20.61	70.8–74.6
ImageNetV2	convnext	5	3.70–28.59	63.7–73.8
	efficientnet	4	4.65–9.11	63.0–66.1
	ghostnet	1	5.18	60.7
	mnasnet	1	4.38	62.2
	mobilenetv2	3	1.97–6.11	51.8–63.9
	mobilenetv3	2	2.54–5.48	53.8–62.8
	regnet	4	5.16–9.19	58.9–64.5
	resnet	4	11.69–44.55	57.0–65.8
	vgg	2	132.87–138.37	57.3–60.3
	vit	4	5.72–22.05	59.3–70.3

5 Results

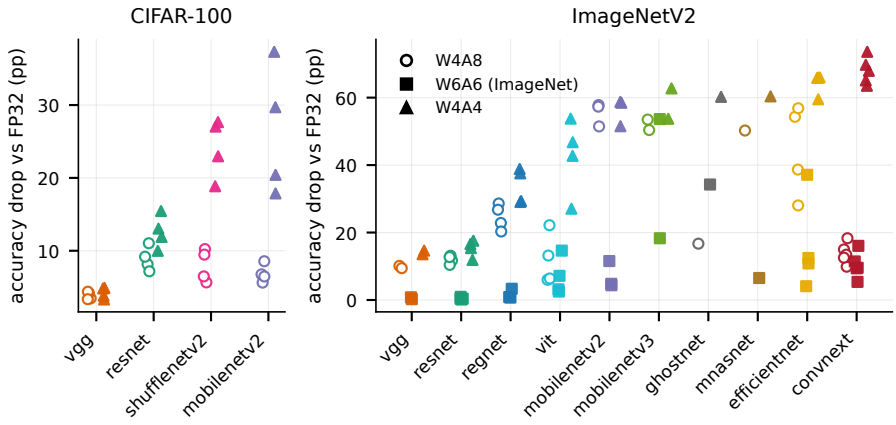
5.1 How unevenly PTQ damages different architectures

Fig. 2 shows the ground truth (per-model values are listed in Tables 5 and 6 in the appendix). W8A8 is nearly lossless (at most 0.35 pp on CIFAR-100; median 0.53 pp on ImageNetV2, with outliers mobilenetv3_small, deit_small and vit_tiny). At W4A8 the drop ranges from 3.4 to 11.0 pp on CIFAR-100 and from 6.0 to 57.8 pp on ImageNetV2, where depthwise-separable families (MobileNetV2, MobileNetV3, MnasNet, EfficientNet) lose 44–56 pp on average against 10–14 pp for ResNet, VGG, ViT and ConvNeXt. W4A4 drops range from 3.3 to 37.3 pp on CIFAR-100, and nearly half of the ImageNet zoo collapses. Calibration variability is small at W4A8 (standard deviation $\leq$ 0.1 pp on CIFAR-100, $\leq$ 0.29 pp on ImageNetV2) and becomes visible only at the strongest settings (up to 2.6 pp at W4A4 on CIFAR-100; up to 1.6 pp at W6A6 and 4.1 pp at W4A4 on ImageNetV2).

5.2 CIFAR-100 (discovery): Amos tracks W4A4 degradation

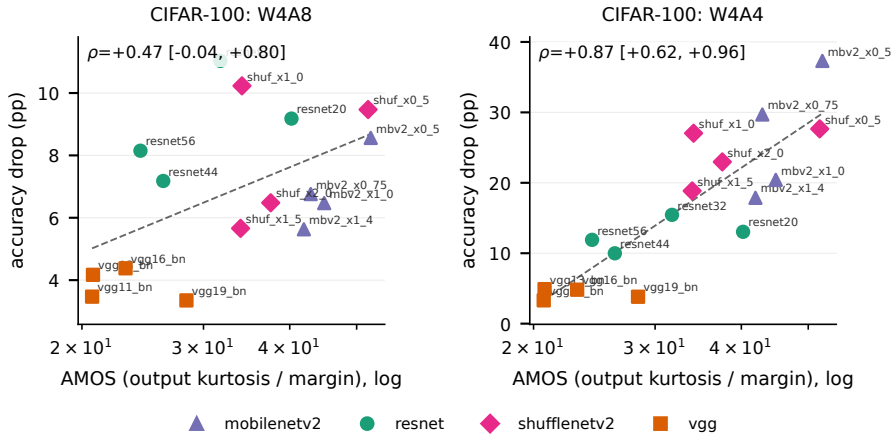
Fig. 3 shows that Amos increases with the W4A4 drop (Spearman $\rho = +0.87$, 95% CI $[0.62, 0.96]$, Kendall $\tau = +0.70$) and is weaker at W4A8 (+0.47, CI $[-0.04, 0.80]$). The W4A4 relation is not an artifact of family or size (Fig. 4): it remains +0.48 within families, +0.83 after removing the effect of parameter count and of the weight-range ratio, and lies in $[+0.77, +0.90]$ when any one family is held out. It is one of the eight variant–target pairs that survive the Holm correction over 60 tests. The input-side variant Amos-in has the opposite sign (−0.78 pooled, +0.22 within families); it ranks architecture families rather than models, and we report it as an ablation.

Held-out prediction. Estimating the W4A4 accuracy of a family not seen during fitting, in terms of mean absolute error, Amos achieves 5.13 pp, while FP32 accuracy alone achieves 12.35 pp and the training mean 10.48 pp. The data-free weight-range ratio performs similarly (4.85 pp), and the two combined perform best (4.32 pp).



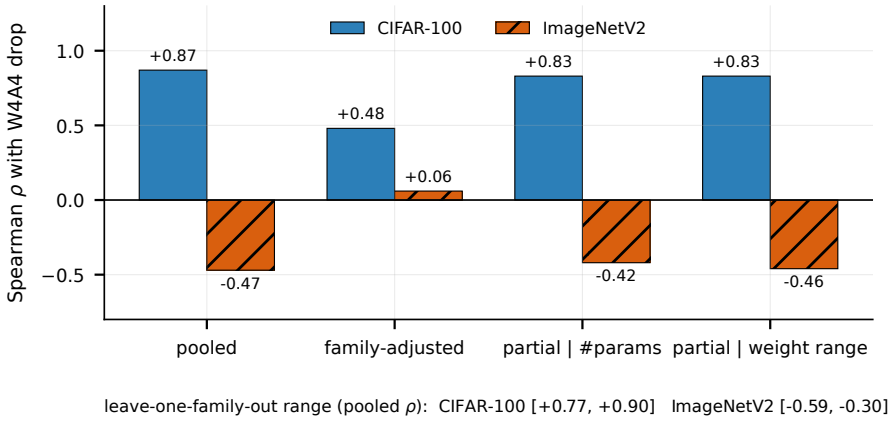
Strip plots of accuracy drop in percentage points by architecture family for CIFAR-100 (left) and ImageNetV2 (right), with separate markers for W4A8, W6A6 and W4A4. VGG and ResNet families have the smallest drops; depthwise-separable families such as MobileNetV2, MobileNetV3 and EfficientNet have the largest, with W4A4 drops above 50 points for many ImageNet models.

Fig. 2. Accuracy drop under PTQ by architecture family. Depthwise-separable families are far more fragile than dense-convolution and transformer families, and W4A4 collapses most ImageNet models.



Two scatter plots for CIFAR-100 with the AMOS score on a logarithmic x-axis and accuracy drop in percentage points on the y-axis, one for W4A8 and one for W4A4. Points are marked by architecture family and labeled by model name. An upward dashed trend line is shown in both; the W4A4 panel is much tighter, with a Spearman correlation of 0.87.

Fig. 3. CIFAR-100: AMOS versus accuracy drop at W4A8 (left) and W4A4 (right). The annotation gives Spearman ρ with its 95% bootstrap CI; the dashed line is a least-squares fit on the log score.



Grouped bar chart of the Spearman correlation between AMOS and the W4A4 accuracy drop under four analyses: pooled, family-adjusted, partial given parameter count, and partial given weight range. CIFAR-100 bars are positive in all four (0.87, 0.48, 0.83, 0.83); ImageNetV2 bars are negative or near zero (-0.47, 0.06, -0.42, -0.46).

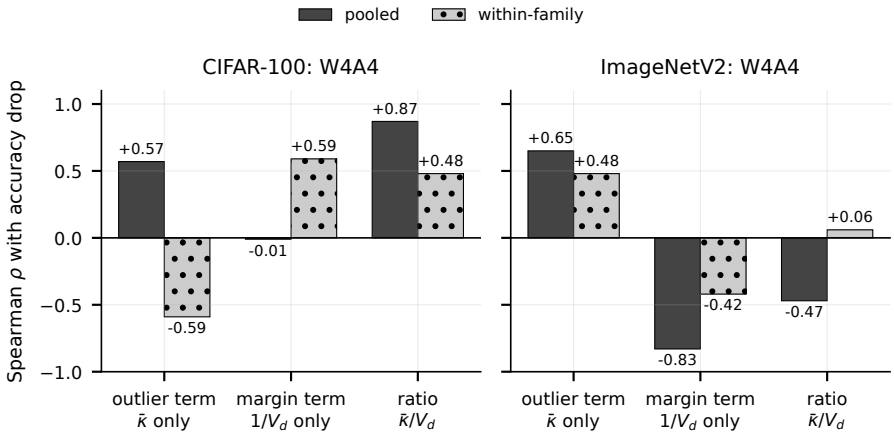
Fig. 4. Controls for AMOS against the W4A4 drop. On CIFAR-100 the correlation survives every control; on ImageNetV2 it does not (Sect. 5.3). The line below the plot gives the range of the pooled correlation when each architecture family is left out in turn.

Which term matters? The ratio is split into its two terms in Fig. 5 and Table 3. On CIFAR-100, the outlier term alone is positively correlated pooled ($\rho = +0.57$) but negatively correlated within families (-0.59), while the margin term alone is uncorrelated pooled (-0.01) but positively correlated within families ($+0.59$). The product combines the between-family information of the first term with the within-family information of the second ($+0.87$ pooled, $+0.48$ within). When the margin term is added, the held-out error is reduced from 6.68 to 5.13 pp. Fig. 6 illustrates how the outlier term works: depthwise-separable networks have spiky layer-output kurtosis profiles, whereas VGG networks have smooth ones. Fig. 13 in the appendix lists all AMOS variants and aggregation choices.

5.3 ImageNetV2 (confirmation): the ratio does not transfer

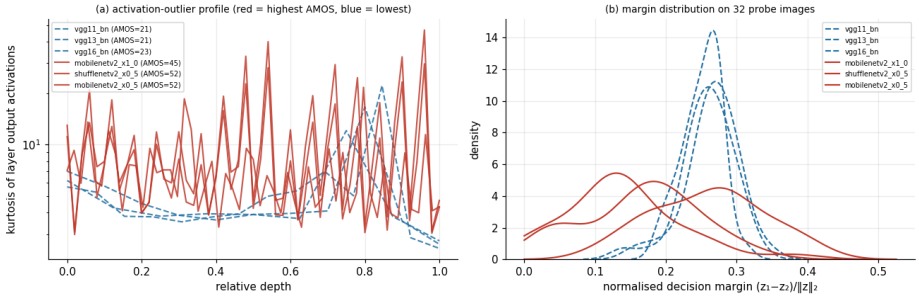
On ImageNetV2, H1 is not supported (Fig. 7): AMOS correlates negatively with the W4A4 drop ($\rho = -0.47$, 95% CI $[-0.70, -0.14]$; family-adjusted $+0.06$) and is not significantly correlated at W6A6 ($\rho = -0.25$, CI $[-0.55, +0.10]$). Fourteen of the 30 models are at the accuracy floor at W4A4, so we consider W6A6 to be the primary test there; the verdict is the same at both settings. All signals are compared across the two datasets in Table 2 and Fig. 8.

The outlier term passes, while the margin term flips. The outlier term alone is found to have a similar size and sign on ImageNetV2 ($\rho = +0.65$ at W4A4; within families:



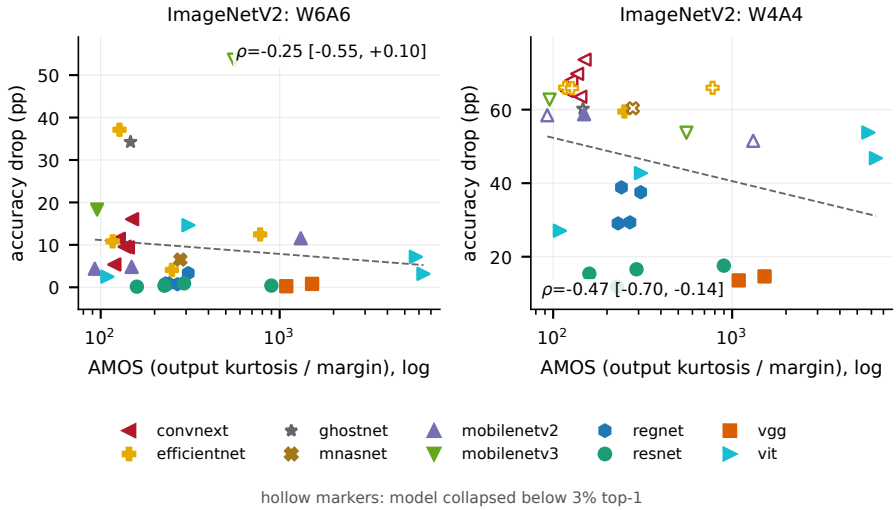
Two bar charts, for CIFAR-100 and ImageNetV2, of pooled and within-family Spearman correlation with the W4A4 accuracy drop for three scores: the outlier term alone, the margin term alone, and their ratio. On CIFAR-100 the ratio is highest (0.87 pooled); on ImageNetV2 the outlier term is positive (0.65 pooled) and the margin term strongly negative (-0.83).

Fig. 5. The two terms of Amos at W4A4, pooled and within architecture families: output-kurtosis term only, $1/V_d$ term only, and their ratio (Amos).



Left: per-layer output kurtosis against relative network depth on a logarithmic axis for six CIFAR-100 models. Dashed lines for the three VGG models stay low and smooth; solid lines for MobileNetV2 and ShuffleNetV2 models show frequent tall spikes. Right: density curves of the normalized logit margin on 32 probe images for the same models; the VGG margins form narrow peaks near 0.27, while the other models are broader and shifted to lower margins.

Fig. 6. Per-layer output-kurtosis profiles (left) and normalized margin distributions (right) of the three lowest- (dashed, VGG-11/13/16) and three highest-Amos (solid, MobileNetV2/ShuffleNetV2) CIFAR-100 models.

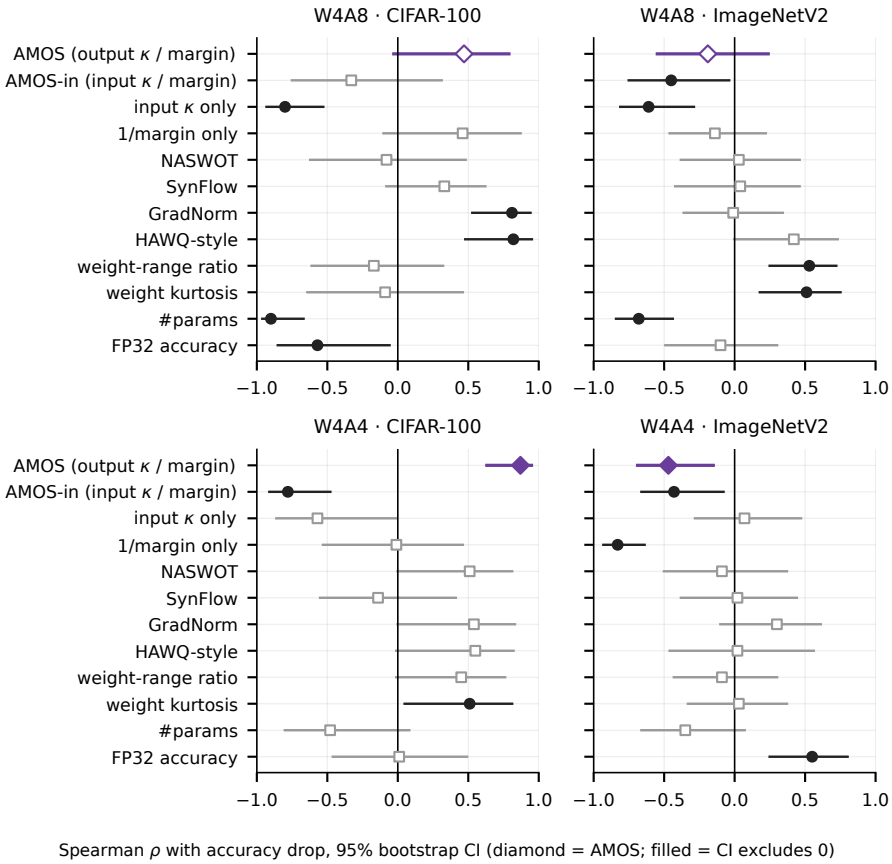


Two scatter plots for ImageNetV2 with the AMOS score on a logarithmic x-axis and accuracy drop on the y-axis, for W6A6 and W4A4, with markers by architecture family. The dashed trend lines slope slightly downward, with Spearman correlations of -0.25 and -0.47. Hollow markers indicate models that collapsed below 3 percent top-1 accuracy.

Fig. 7. ImageNetV2: AMOS versus accuracy drop at W6A6 (left) and W4A4 (right). The CIFAR-100 relation does not carry over. Hollow markers: models collapsed below 3% top-1.

Table 2. Spearman ρ between each zero-shot signal and the accuracy drop under PTQ, per dataset and bit-width. **Bold**: 95% bootstrap CI excludes zero. Full CIs in Fig. 8; W6A6 was evaluated on ImageNetV2 only.

	W4A8		W4A4		W6A6
Signal	CIFAR	ImgNet	CIFAR	ImgNet	ImgNet
AMOS (output κ / margin)	+0.47	−0.19	**+0.87**	**−0.47**	−0.25
AMOS-in (input κ / margin)	−0.33	**−0.45**	**−0.78**	**−0.43**	−0.32
input κ only	**−0.80**	**−0.61**	−0.57	+0.07	−0.00
1/margin only	+0.46	−0.14	−0.01	**−0.83**	**−0.64**
NASWOT	−0.08	+0.03	+0.51	−0.09	**−0.46**
SynFlow	+0.33	+0.04	−0.14	+0.02	−0.18
GradNorm	**+0.81**	−0.01	+0.54	+0.30	+0.09
HAWQ-style	**+0.82**	+0.42	+0.55	+0.02	+0.17
weight-range ratio	−0.17	**+0.53**	+0.45	−0.09	+0.14
weight kurtosis	−0.09	**+0.51**	**+0.51**	+0.03	+0.14
#params	**−0.90**	**−0.68**	−0.48	−0.35	**−0.46**
FP32 accuracy	**−0.57**	−0.10	+0.01	**+0.55**	+0.23



Four forest plots, each listing twelve zero-shot signals with their Spearman correlation and 95 percent bootstrap interval against accuracy drop: W4A8 and W4A4 on CIFAR-100 (left) and on ImageNetV2 (right). AMOS is drawn as a diamond. Parameter count is strongly negative at W4A8 on both datasets; AMOS is positive on CIFAR-100 at W4A4 and negative on ImageNetV2.

Fig. 8. Correlation of every signal with the accuracy drop, with 95% bootstrap intervals, at W4A8 (top row) and W4A4 (bottom row) on CIFAR-100 (left) and ImageNetV2 (right).

Table 3. Decomposition of AMOS at W4A4 (Spearman ρ with accuracy drop): pooled over all models, within architecture families, and after removing the effect of parameter count or weight-range ratio. The outlier term has a positive pooled correlation and positive partial correlations (given parameter count and given weight range) on both datasets; the margin term does not transfer.

Dataset	Signal	Pooled	Within-family	| #params	| weight range
CIFAR-100	output κ only	+0.57	−0.59	+0.89	+0.38
	1/margin only	−0.01	+0.59	−0.47	+0.34
	AMOS (ratio)	+0.87	+0.48	+0.83	+0.83
ImageNetV2	output κ only	+0.65	+0.48	+0.67	+0.65
	1/margin only	−0.83	−0.42	−0.84	−0.85
	AMOS (ratio)	−0.47	+0.06	−0.42	−0.46

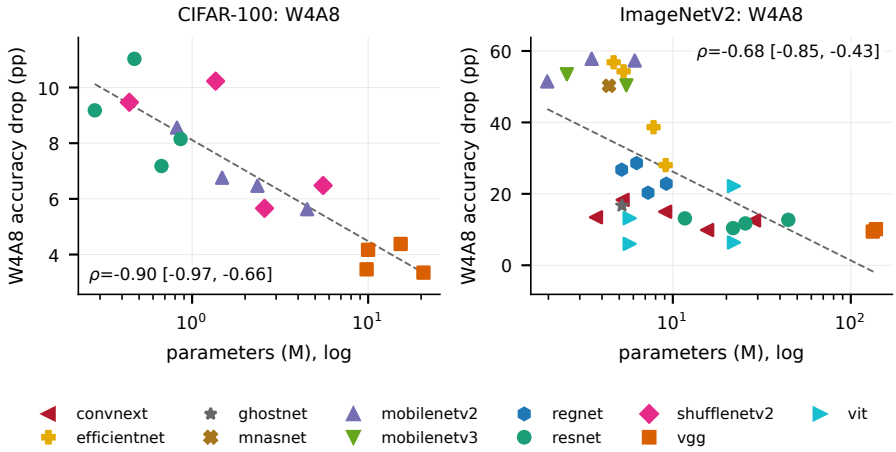
+0.48; given parameter count: +0.67; given weight range: +0.65) as on CIFAR-100 (+0.57; +0.89 and +0.38). The margin term, however, has a negative correlation of −0.83 (95% CI $[-0.94, -0.63]$) with the W4A4 drop on ImageNetV2, whereas it has a correlation of −0.01 (95% CI $[-0.54, +0.47]$) with the W4A4 drop on CIFAR-100. Since the two datasets use different training recipes (one for CIFAR-100 and many `timm` recipes for ImageNet), the margin seems to contain a recipe-dependent logit scale, which dominates the ratio on ImageNet. We consider the outlier-term result an exploratory component analysis of a pre-declared score and provide only point estimates.

5.4 What replicates across datasets

Model size is a predictor of W4A8 loss. Parameter count has $\rho = -0.90$ (CI $[-0.97, -0.66]$) on CIFAR-100 and −0.68 (CI $[-0.85, -0.43]$) on ImageNetV2, with drop-one-family ranges of $[-0.94, -0.79]$ and $[-0.76, -0.60]$, respectively (Fig. 9). It is also the best held-out predictor of W4A8 accuracy on CIFAR-100 (1.54 pp MAE). Signals that look strong at W4A8 on CIFAR-100 are mostly size-dependent: GradNorm and the HAWQ-style score are +0.81 and +0.82, respectively, but drop to +0.01 and −0.25 given parameter count. Input κ alone (−0.80 and −0.61 on the two datasets) drops to +0.19 and −0.30 given parameter count.

FP32 accuracy mis-selects at W4A4. The selection regret of choosing by FP32 accuracy is 11.6 pp on CIFAR-100 (random: 14.2 pp) and 48.4 pp on ImageNetV2 (random: 32.1 pp; Fig. 10). The regret is lowered to 0.0 and 5.9 pp by adding the number of parameters, and to 0.3 pp by adding Amos on CIFAR-100. On the edge-sized pool of 10 CIFAR-100 models (≤3 M parameters), Amos improves regret from 17.3 pp (FP32 accuracy only) to 0.95 pp, and the leave-one-out rank correlation is +0.66 versus −0.76 for FP32 accuracy alone. Only parameter count (10.9 pp) and the HAWQ-style score (21.1 pp) do better than FP32 accuracy alone (32.0 pp) on the corresponding ImageNetV2 pool (13 models, ≤6 M).

For held-out prediction (Fig. 11), at W6A6 on ImageNetV2 the selection regret of all methods is between 0 and 9.5 pp, and the HAWQ-style score (5.75 pp MAE) and the weight-range ratio (6.31 pp) are below the training-mean baseline (9.24 pp), while FP32 accuracy (10.04 pp) and Amos (10.52 pp) are not. At W4A4 on ImageNetV2 the



Two scatter plots of parameter count in millions on a logarithmic x-axis against W4A8 accuracy drop for CIFAR-100 and ImageNetV2, with markers by architecture family. Larger models have smaller drops on both datasets; the Spearman correlations are -0.90 and -0.68.

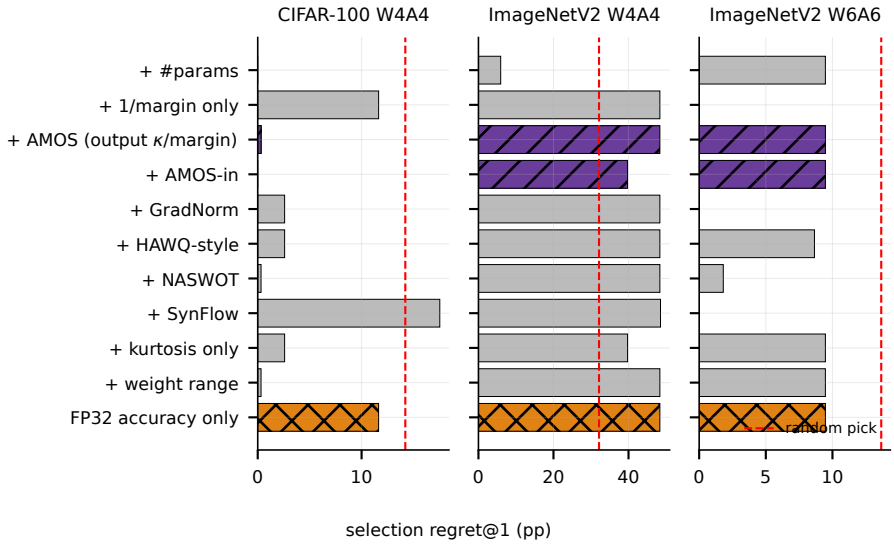
Fig. 9. Parameter count versus W4A8 accuracy drop on both datasets; the annotation gives Spearman ρ with its 95% bootstrap CI.

training-mean error is 19.3 pp and is improved only by parameter count (15.9 pp), the margin term (16.0 pp) and Amos-in (17.6 pp).

5.5 Cost and stability of the measurement

The other forward/backward proxies require a comparable amount of time to computing Amos on CIFAR-100 (0.057 s) and ImageNetV2 (0.21 s), while evaluating one bit-width with three calibration subsets requires 8.5 s and 36.6 s, respectively (Table 4). The median per-model ratio of the two is 201× on CIFAR-100 and 175× on ImageNetV2. The model ranking is consistent across different probe batches (mean pairwise $\rho = 0.988$ on CIFAR-100 and 0.953 on ImageNetV2) and remains similar from $N = 8$ ($\rho = 0.86$) to $N = 128$ ($\rho = 0.88$) probe images on CIFAR-100 (Fig. 12). It is important to note that stable measurement does not necessarily mean predictive validity, so Sects. 5.2–5.4 assess the latter separately.

Exploratory observation of the type of probe. With a Gaussian-noise probe, Amos correlates positively with the W4A4 drop on both datasets (+0.23 on CIFAR-100 and +0.55 on ImageNetV2; +0.46 at W6A6), whereas real-image probes give +0.87 and −0.47. This is a post hoc observation based on three probe batches and without confidence intervals, and it suggests that the structural (data-independent) part of the score might be the more transferable one. We put it forward as a hypothesis for further research.

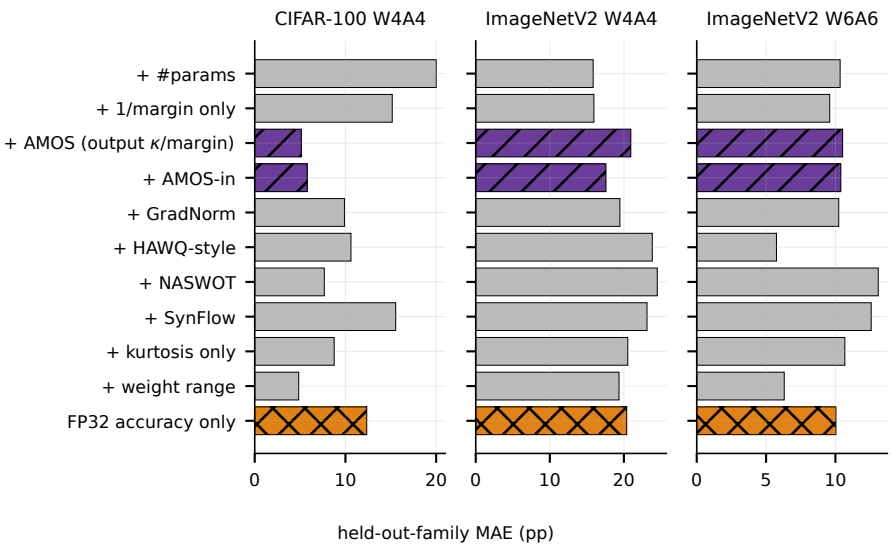


Three horizontal bar charts of selection regret at one pick, in percentage points, for ten selection rules and a dashed random-pick reference line: CIFAR-100 W4A4, ImageNetV2 W4A4 and ImageNetV2 W6A6. FP32 accuracy only has a regret of 11.6 and 48.4 points in the first two; adding parameter count lowers it to about 0 and 5.9 points.

Fig. 10. Selection regret@1 (lower is better). Crossed bars: FP32 accuracy only; hatched purple bars: Amos variants; dashed line: expected regret of a random pick.

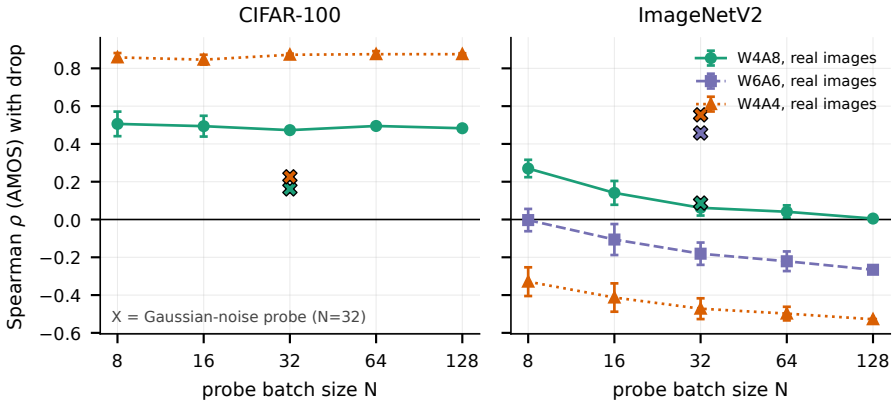
Table 4. Median wall-clock time per model on a Tesla T4 (seconds). PTQ times include calibration, quantization and full evaluation (10,000 images on CIFAR-100, 5,000 on ImageNetV2). The speed-up is the median over models of the per-model ratio of the three-subset W4A8 time to the Amos time, so it need not equal the ratio of the medians shown.

	CIFAR-100	ImageNetV2
Amos (one forward pass, 32 images)	0.057	0.21
NASWOT	0.040	0.36
SynFlow	0.067	0.13
Gradient norm	0.064	0.13
HAWQ-style (Hutchinson)	1.0	4.5
One PTQ run (W4A8, one calibration subset)	2.8	12.2
Three calibration subsets (W4A8)	8.5	36.6
All bit-widths × three subsets	25.1	146
Median speed-up of Amos vs. three subsets	201×	175×



Three horizontal bar charts of leave-one-family-out mean absolute error, in percentage points, for ten predictors and FP32 accuracy only: CIFAR-100 W4A4, ImageNetV2 W4A4 and ImageNetV2 W6A6. On CIFAR-100 the AMOS variants and weight range are lowest (about 5 points); on ImageNetV2 W4A4 most predictors are around 20 points.

Fig. 11. Leave-one-family-out error in predicted quantized accuracy (pp). "+ X" denotes FP32 accuracy plus signal X; hatched purple bars: Amos variants; crossed bars: FP32 accuracy only.



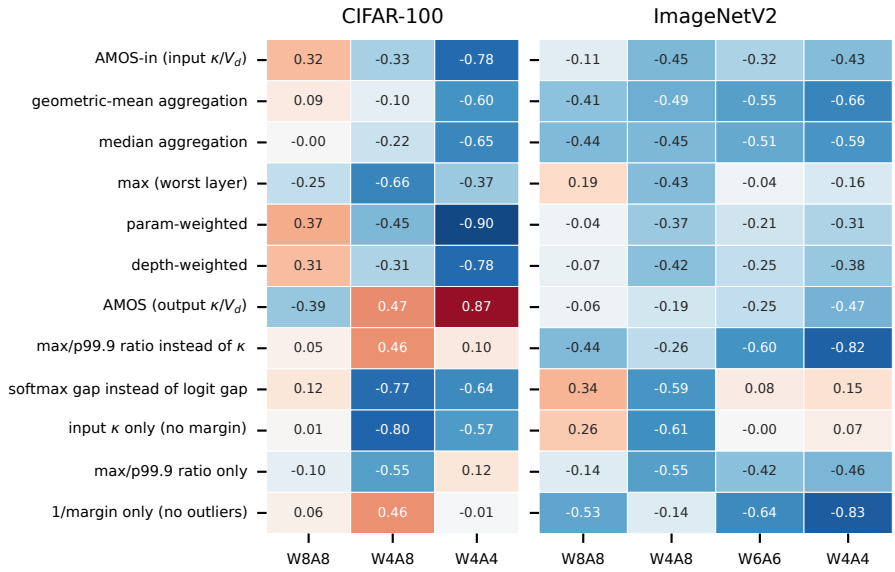
Two line charts of the Spearman correlation between the output-kurtosis AMOS score and accuracy drop against probe batch size from 8 to 128 images, for CIFAR-100 (left) and ImageNetV2 (right), with separate marker shapes and line styles per bit-width setting. Curves are flat across batch sizes. X markers at 32 images show the Gaussian-noise probe: positive correlations on both datasets.

Fig. 12. Effect of probe size and probe type on the correlation between AMOS (output-kurtosis version) and the accuracy drop (mean ± s.d. over three independent probe batches). X marks a probe of Gaussian noise instead of real images.

6 Discussion

The results of the audit. A cheap statistic can be impressive on a small zoo and yet fail on ImageNetV2: the CIFAR-100 result for AMOS is strong across all of our controls, but it is reversed on ImageNetV2. This was made apparent by two practices: testing on a second dataset with an a priori hypothesis and decomposing the score. The decomposition concentrates the failure in the margin term, which depends on the training recipe, while the outlier term does not change sign between datasets. For practical use, we draw three conclusions: at W4A8, model size is a sufficient and robust guide; at W4A4, FP32 ranking should not be trusted as a guide for selection; and, for aggressive settings, an output-outlier screen is a cheap complement to, but not a replacement for, a quantization run.

Scope. The ground truth is a simulated PTQ pipeline with per-tensor activation quantization and without learned rounding, which makes 4-bit activation quantization hard for depthwise and transformer models. More powerful PTQ methods would cause absolute drops to change and may result in different rankings. The CIFAR-100 zoo consists of four families, and the ImageNet zoo is a mixture of training recipes, so the confidence intervals reflect only model-level sampling. ImageNetV2 is not the original validation set but a re-collected test set.



Two heatmaps of the Spearman correlation between twelve variants of the AMOS score and the accuracy drop, with one column per bit-width setting: three for CIFAR-100 and four for ImageNetV2. Cell values are printed. Output-kurtosis AMOS is strongly positive for CIFAR-100 W4A4 (0.87) and negative on ImageNetV2.

Fig. 13. Spearman ρ of every AMOS variant with the accuracy drop: aggregation over layers, input versus output kurtosis, ratio of maximum to 99.9th-percentile activation, softmax gap instead of logit gap, and each term alone. "Input κ only" uses layer inputs; the "outlier term" of Fig. 5 uses layer outputs.

7 Conclusion

Across 46 pretrained models, 4-bit PTQ fragility follows architecture family and size more than any single cheap signal. A one-pass score combining output-activation outliers and logit margin is strongly predictive in-distribution, but only its outlier component transfers to ImageNetV2, and model size remains the most consistent predictor of W4A8 loss. The audit protocol and code are available at `[REPOSITORY URL TO BE ADDED]`, so that quantization proxies can be validated across datasets before they are used for model selection. Natural next steps are intervals for the outlier-only score on larger zoos, stronger PTQ baselines, per-channel activation quantization, and the data-free noise-probe variant.

A Additional Results

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Table 5. Per-model results on CIFAR-100, sorted by family and size. AMOS is the output-kurtosis score; Params in millions, FP32 is the top-1 accuracy (%); the bit-width columns give the drop (FP32 minus PTQ accuracy, pp), averaged over 3 calibration subsets.

Model	Params	FP32	Amos	W4A8	W4A4
mobilenetv2_x0_5	0.82	71.2	52	8.6	37.3
mobilenetv2_x0_75	1.48	74.1	43	6.8	29.7
mobilenetv2_x1_0	2.35	74.3	45	6.5	20.4
mobilenetv2_x1_4	4.50	76.3	42	5.6	17.9
resnet20	0.28	68.8	40	9.2	13.0
resnet32	0.47	70.1	32	11.0	15.4
resnet44	0.67	71.7	26	7.2	10.0
resnet56	0.86	72.6	24	8.2	11.9
shufflenetv2_x0_5	0.44	67.8	52	9.5	27.7
shufflenetv2_x1_0	1.36	72.6	34	10.2	27.0
shufflenetv2_x1_5	2.58	74.2	34	5.7	18.9
shufflenetv2_x2_0	5.55	75.5	38	6.5	23.0
vgg11_bn	9.80	70.8	21	3.5	3.3
vgg13_bn	9.99	74.6	21	4.2	4.9
vgg16_bn	15.30	74.0	23	4.4	4.8
vgg19_bn	20.61	73.9	28	3.4	3.8

Table 6. Per-model results on ImageNetV2 (5,000 evaluation images; sorted by family and size). Columns as in Table 5.

Model	Params	FP32	Amos	W4A8	W6A6	W4A4
convnext_atto	3.70	63.7	142	13.4	9.4	63.5
convnext_femto	5.22	65.3	119	18.3	5.4	65.1
convnext_pico	9.05	68.1	126	15.0	11.4	67.9
convnext_nano	15.59	69.9	136	9.9	9.6	69.7
convnext_tiny	28.59	73.8	150	12.6	16.1	73.6
efficientnet_lite0	4.65	63.0	250	56.9	4.1	59.5
efficientnet_b0	5.29	65.9	127	54.3	37.1	65.8
efficientnet_b1	7.79	66.0	778	38.7	12.5	65.9
efficientnet_b2	9.11	66.1	117	28.0	10.8	66.0
ghostnet_100	5.18	60.7	147	16.7	34.2	60.2
mnasnet_100	4.38	62.2	279	50.2	6.5	60.4
mobilenetv2_050	1.97	51.8	1311	51.5	11.6	51.5
mobilenetv2_100	3.50	60.2	93	57.8	4.4	58.5
mobilenetv2_140	6.11	63.9	149	57.3	4.8	58.8
mobilenetv3_small_100	2.54	53.8	556	53.5	53.6	53.7
mobilenetv3_large_100	5.48	62.8	95	50.4	18.3	62.7
regnetx_004	5.16	58.9	308	26.8	3.3	37.5
regnety_008	6.26	64.0	241	28.6	0.9	38.9
regnetx_008	7.26	62.4	269	20.3	0.7	29.3
regnetx_016	9.19	64.5	230	22.8	0.9	29.1
resnet18	11.69	57.0	292	13.1	0.9	16.6
resnet34	21.80	60.5	227	10.4	0.4	11.9
resnet50	25.56	63.1	159	11.7	0.1	15.4
resnet101	44.55	65.8	899	12.7	0.4	17.6
vgg11_bn	132.87	57.3	1517	9.4	0.8	14.7
vgg16_bn	138.37	60.3	1088	10.1	0.2	13.6
deit_tiny	5.72	59.3	109	6.0	2.5	27.1
vit_tiny	5.72	62.6	5767	13.2	7.2	53.8
deit_small	22.05	68.2	309	22.2	14.6	42.7
vit_small	22.05	70.3	6357	6.4	3.2	46.8

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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